

Over Pessimistic anticipation in Response to GenAI. DiD/EventStudy Analysis on the Occupational Exposure to GenAI

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1 Introduction

When ChatGPT was released in November 2022, it became possible, to point to a single moment when generative AI entered everyday use. I ask two simple questions: *How did occupations with high AI exposure respond economically? How much of the response was anticipation of harm rather than realized economic harm?* The main outcome of interest is the rent paid by workers in each occupation—a forward looking measure of how secure people feel about their incomes—alongside wages and employment.¹

The key idea is that generative AI does not affect all jobs equally. Some occupations consist largely of tasks a language model can now do; others barely overlap with it. If AI matters, the gap between these two groups should widen after late 2022, and not before. We measure that widening directly.

2 Data

We combine two ingredients.

Workers. Annual American Community Survey (ACS) microdata for 2017–2024 provide a large, representative sample of US residents, including their occupation, earnings, rent, hours, education, and recent moves.

Exposure. Each occupation is assigned a single generative-AI exposure score, measuring how much of its work overlaps with what these tools can do. Taken from **Eisfeldt et al (2023)**.

3 Building the panel

We collapse the millions of individual records into one row per *occupation* $\times$ *year*. Of the occupations we can score for AI exposure, **351** are observed in every year from 2017 to 2024; these form the balanced panel used throughout, a total of $351 \times 8 = 2,808$ observations. Table 1 describes them, and Table 2 compares the most- and least-exposed occupations before ChatGPT.

¹This methodology obviously needs changing. See Section 8

Table 1: Summary statistics for the balanced occupation×year panel, 2017–2024.

Variable	N	Mean	SD	Median	Max
AI exposure score	2808	0.250	0.226	0.179	0.958
Employment (persons)	2808	204,393	357,630	72,401	3,004,241
Employment share	2808	0.002	0.004	0.001	0.033
Unemployment rate	2808	0.040	0.037	0.030	0.419
Mean wage (\$)	2808	61,422	29,791	54,585	222,215
Mean hours/week	2808	40.9	3.8	41.3	56.6
Mean rent (\$/mo)	2808	1,381	350	1,327	3,085
Renter share	2808	0.341	0.109	0.328	0.794
Share with BA+	2808	0.384	0.306	0.282	1.000
Share female	2808	0.410	0.280	0.399	0.980
Mean age	2808	39.4	1.8	39.4	46.2
Work-from-home share	2808	0.115	0.130	0.057	0.720
Migration rate	2808	0.143	0.039	0.140	0.370

One observation per occupation×year over the 351 occupations present in all 8 years. AI exposure is fixed at its 2021 value.

4 Defining who is “treated”

Exposure is fixed at its **2021 value**, just before ChatGPT’s public released. We use three complementary ways of splitting occupations, and report all three:

- **Extreme groups.** Sort occupations into thirds by exposure and compare the most-exposed third (*high*) against the least-exposed third (*low*).
- **Matched groups.** Compare the most-exposed occupations against an equal number of occupations with *zero* exposure, so the two sides are balanced in size.
- **Continuous dose.** Use the full exposure score, standardised so that a one-unit change equals one standard deviation across occupations.

5 The difference-in-differences estimate

For an outcome y in occupation i and year t we estimate

$$y_{it} = \beta (\text{Exposure}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where:

- Exposure_i is the high-group indicator (extreme design) or the standardised score (dose design), *fixed at 2021*;
- $\text{Post}_t = 1$ from 2023 onward—the first survey year in which a ChatGPT effect could appear;
- α_i are **occupation fixed effects**, absorbing every permanent difference between occupations (a lawyer always earns more than a cashier);
- δ_t are **year fixed effects**, absorbing everything that moves all occupations together in a given year (inflation, the business cycle).

Table 2: Baseline (2021) characteristics of the most- vs least-exposed occupation terciles.

Variable	Low exposure	High exposure	Difference
Employment (persons)	157,292	205,936	48,644
Employment share	0.002	0.002	0.001
Unemployment rate	0.067	0.040	−0.028
Mean wage (\$)	49,903	75,106	25,203
Mean hours/week	40.4	41.1	0.7
Mean rent (\$/mo)	1,212	1,534	322
Renter share	0.352	0.309	−0.044
Share with BA+	0.208	0.582	0.374
Share female	0.315	0.536	0.221
Mean age	39.5	39.4	−0.1
Work-from-home share	0.073	0.362	0.288
Migration rate	0.133	0.156	0.022

Group means across occupations in each tercile, baseline year 2021. Low: 151 occupations; high: 110.

6 Event study

We replace the single Post_t term with one coefficient per year, measured relative to **2022**, the last year before ChatGPT:

$$y_{it} = \sum_{t \neq 2022} \beta_t (\text{Exposure}_i \times \mathbf{1}[\text{year} = t]) + \alpha_i + \delta_t + \varepsilon_{it}. \quad (2)$$

7 Outcome Variable: Rent

Exposed occupations saw rent fall after ChatGPT.² The most-exposed occupations paid about **1.9% less** than those in the least-exposed occupations (extreme design). The continuous-dose design tells the same story: each one-standard-deviation increase in exposure lowers rent by about **0.8%**, see Table 3. The event study confirms parallel trends assumption for all three groups, and sees a rent gap open after treatment, see Figure 2.

Table 3: Effect of GenAI exposure on log rent (pooled difference-in-differences).

Specification	Extreme (high – low)	Dose (per SD)
Baseline	−0.019***	−0.008***
+ renter share	−0.019***	−0.008***
+ wages	−0.016**	−0.007**
+ employment	−0.019***	−0.008***
+ all three	−0.016**	−0.007**

Standard errors clustered by occupation. Occupation and year fixed effects; weighted by 2021 employment.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

²Rent is measured only among the employed. See Section 8

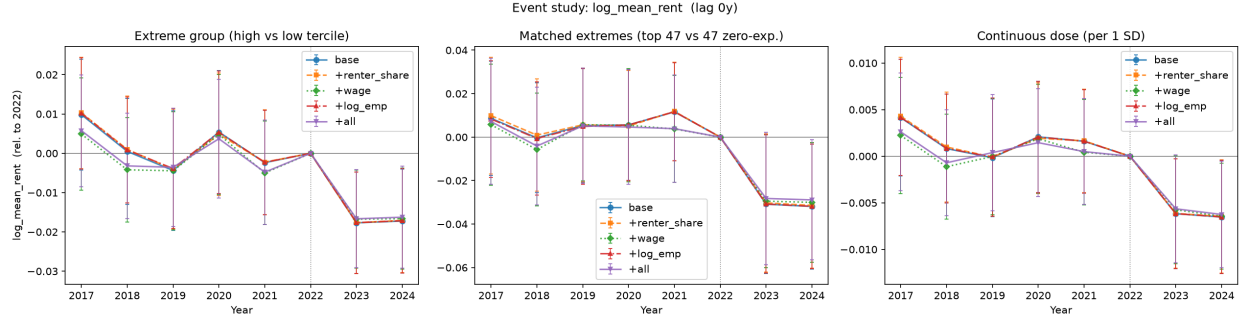


Figure 1: Event study for log rent.

Each point is the rent gap between exposed and unexposed occupations in that year, measured relative to 2022.

Bars are 95% confidence intervals.

7.1 Discussion of controls

We add three controls, each ruling out a reason the rent fall could be misleading rather than real. We control for **renter share** in case exposed renters bought homes, which would change who is left renting. We control for **wages** in case rent simply fell because pay fell. We control for **employment** in case rent dropped only because people lost their jobs. Together, these separate the anticipation response from concrete changes in income or jobs.³

7.2 Stratified test: Remote work

The COVID-19 pandemic let some high-exposure office workers move to cheaper areas, which could lower their rent for reasons unrelated to AI—the “donut” effect. To test this we split occupations by whether the job can be done from home and re-estimate the rent effect separately within each group (Table 4).

The decline is fully present among jobs that **cannot** be done remotely (−1.9%, significant) and absent among jobs that **can** (≈ 0 , not significant).⁴ If remote work were the cause the pattern would be the reverse—an effect where people can actually relocate. Because the decline appears exactly where remote work is impossible, it is not the donut effect.

Table 4: Rent effect split by whether the job can be done from home.

Group	Extreme (high – low)	Dose (per SD)
Cannot work from home	−0.019***	−0.018**
Can work from home	+0.001	−0.001

Outcome is log rent. Standard errors clustered by occupation; occupation and year fixed effects; weighted by 2021 employment. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7.3 Placebo test

To ensure the rent fall is specific to people still exposed to the labour market, we repeat the analysis on retired renters (out of the labour force, aged 60+), grouped by their former occupation.

³See Section 8 for discussions on additional controls

⁴More future work will be on this, see Section 8

As expected, there is no effect (Table 5): both estimates are small and insignificant.

Table 5: Placebo: rent of retired renters (out of the labour force, 60+), by former occupation.

Outcome	Extreme (high – low)	Dose (per SD)
Retiree rent (log)	+0.026	+0.010

Neither estimate is significant. Standard errors clustered by occupation; occupation and year fixed effects; weighted by 2021 employment.

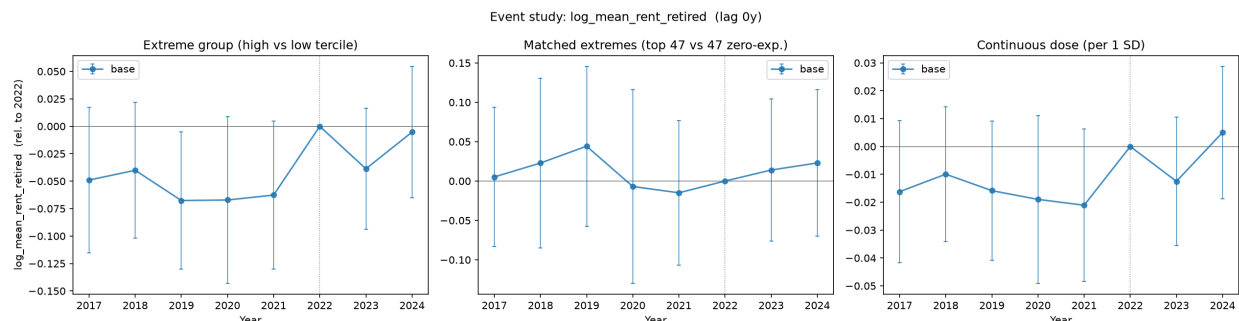


Figure 2: Event study for log rent on retired renters.

Each point is the rent gap between exposed and unexposed occupations in that year, measured relative to 2022. Bars are 95% confidence intervals.

8 What's next

There are obviously issues to be addressed.

- 1. Rent is a noisy signal.** For many people housing is the last thing they cut, before eating out and the rest. We use rent only because **the pattern turned up by chance**. Future work on this will focus on getting other spending data to better investigate the anticipation effect.
- 2. No effect for work-from-home jobs.** We hypothesize two possible reasons for the insignificant results for high WFH groups: these workers already moved during the pandemic recovery, or that WFH workers are (probably) more skilled and therefore have higher pay, thus the financial buffer to pay the same rent. Future work should test both, and check that work-from-home really does track higher pay.
- 3. Migration.** Moving across states is worth studying on its own. It shapes rent, but AI may also drive it: if a job shrinks, workers may have to leave the state where it was based.
- 4. Wages and jobs barely move.** Wages and employment show no real response to ChatGPT. Unemployment ticks up, but its earlier path is too messy to blame on AI. None of this hurts the rent story, which only concerns people who are working.